Echo chambers in BlueSky Starter Packs

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Abstract

As decentralized platforms such as Bluesky continue to grow, individuals are now, more than ever, in charge of what they see and consume with minimal algorithmic influence. This project investigates whether BlueSky Starter Packs promote social diversity or curate echo chambers by focusing on people and feeds listed within. Using a provided dataset of 300,000 Starter Pack URIs and Bluesky's API endpoints we analyze user overlap, thematic categories, and starter pack popularity. We find that there are clusters that exist among starter packs, there are 19 distinct topical groups such as: art, sport, gaming, and news, and individual accounts with high median follower counts are central hubs within communities. Our findings suggest that manual and customizable features have the potential to create echo chambers by siloing users into "social bubbles."

Introduction

Social media is ever-present in the daily lives of millions of people. From MySpace and Facebook to the new emerging platforms like Threads and BlueSky, access to and consumption of information grows exponentially everyday (Cinelli et al., 2021). However, like with many social platforms, there are possibilities of reinforced perspectives or amplified voices of the vocal minority (Vosoughi et al., 2018; Wu & Resnick, 2021). This phenomenon is also referred to as an echo chamber. According to Cabianca, Hammond, and Gutierrez (2020), the definition of "a social media echo chamber is when one experiences a biased, tailored media experience that eliminates opposing viewpoints and differing voices." For our purposes, we narrow down defined echo chambers as 1.) mass exposure to the same or similar people, 2.) exposure to the same or similar content, 3.) exposure to influential perspectives within tightly connected communities. Using this definition, the goal of this research is to identify whether BlueSky starter packs promote social diversity or create echo-chambers.

Bluesky starter packs are curated lists of accounts and feeds—up to 150 people and up to 3 custom feeds—designed to help users discover communities and content when joining the platform. Because Bluesky is a decentralized platform and does not rely on traditional algorithmic recommendation systems to guide content discovery, users must depend more on manual or curated tools to navigate the network. As a result, starter packs play a critical role in shaping user discovery and social network formation.

Our motivation for this project stems from "Bootstrapping Social Networks: Lessons from Bluesky Starter Packs" by Balduf et al. (2025). Starter packs often circulate within communities, creating clusters or "social bubbles" where users promote and follow others within the same social groups. This suggests that starter packs may reinforce community structures and amplify the visibility of certain accounts. Thus creating echo chambers.

Research Questions

Through this project, we aim to answer the following research questions:

- **RQ1** User Overlap: Do different starter packs share overlapping accounts that indicate potential community structures or diversity among packs?
- **RQ2** Thematic Categorization: Do starter pack descriptions reveal distinct themes (journalism, sports, politics, art, tech)?
- **RQ3** Pack Popularity: Do different communities contain accounts with different follower influence levels?

To begin answering these questions, we started by extracting BlueSky Starter Pack (SP) data provided to us by postdoctoral scholar, Alexandros Efstratiou. The dataset contained 300,000 rows of SP URIs, which we scraped using BlueSky's API endpoints: get_starter_pack and get_list. For the creator's information, the necessary SP data, and all feeds and accounts contained within.

Hypotheses

H1: Starter packs will have overlapping accounts that demonstrate echo chambers where people will form a community within similar content.

H2: If the starter pack's description is very specific, we can cluster and categorize them into a specific community using HDBSCAN.

H3: Communities with greater median followers will have more influence

Data Collection

Part 1: Collect SP Metadata

To process the metadata set, we developed a python-based data collection script that extracts relevant information from the SP URIs through Bluesky's API. We partitioned the 300,000 dataset into approximately 65k URIs per person, and each person ran their assigned subset to extract the relevant information.

With the JSONL file containing all URIs parsed using the python scripts, we extracted the information needed with its respective APIs and created several functions to collect data such as get_account_info, get_all_accounts, and get_starter_pack_info.

With the URIs we call getStartPacks from BlueSky API:

- SP DID: unique ID for this starter pack record
- SP Creator DID: unique ID of the user who created the pack
- SP Creator Handle: The username of users who created the SP
- SP Description: Text description of each SP

Finally, we did an initial high level analysis regarding what feeds occur the most and which account has the highest frequency across all SP.

Part 2: Account Information

From the list of account objects contained in each starter pack, we extracted key identity information for each account. Specifically, we recorded the Account DID, which serves as the unique identifier for the account, and the Account Handle, representing the user's public username (e.g., nytimes.com).

Additionally, we tried to fetch the follower count and post counts of the people within the SPs calling the getProfiles API. Within the 65,000 URI, there were 2 million rows of unique pairs of SPs and accounts. Then within the 2 million unique rows, there were 661,635 unique accounts. After conducting the account followers, we computed the median per starter pack:

Median Followers	SP Creator Handle
1343452.0	ottawajames.bsky.social
1323274.5	spincitysj95121.bksy.social
1315450.0	nayandubey.bsky.social
1298331.0	jwrhb.bsky.social
1298331.0	realranchbabe.bsky.social

Table 1: Median followers in starter packs.

Part 3: Categorizing Starter Pack Themes

To conduct the thematic analysis, we began by filtering out through 1,000 SPs that did not have a description then slowly scaled up to 65,000. After filtering, we were left with 20,525 unique SPs. After identifying the different themes of the SP, we used community labels to identify which community had the most SPs and which communities were the most influential such as communities with the highest median followers.

Themes	**# of Themes within 1,000 SP**
Unknown	132
Art	61
Animals	54
Tech	30
Politics	14

Table 2: Starter pack classification using 1,000 SP

Main Analyses

User Overlap and Community Structure

Our first research question that we want to answer is whether different SP share overlapping accounts that indicate potential echochamber-like community structure. To tackle this, we constructed a network graph using NetworkX's community detection (greedy modularity) algorithm, in which each node represents a starter pack and edges connect two SPs if they share at least one account. The edge weight between two nodes is calculated using Jaccard similarity, which is defined as:

$$J(A,\ B)\ =\ \frac{|A \cap B|}{|A \cup B|}$$

where A and B represent the sets of accounts contained in two SPs. Jaccard Similarity measures the proportion of shared accounts relative to the total unique accounts contained in both packs. This means that a higher weight would indicate a greater degree of overlap between packs. Our reasoning for using Jaccard Similarity is that since starter packs vary in size, it can help normalize overlaps relative to the total number of accounts. We also set the similarity threshold to 0.15 to remove weak overlaps and reduce noise in the network. As we had a lot of data, we didn't want to overpopulate the graph with too many nodes, so we decided to focus on the largest connected component to analyze the most meaningful relationship between packs while excluding isolated or weakly connected ones.

The network visualization reveals several clusters of starter packs with high user overlap. In this context, clusters refer to the groups of nodes identified by the community detection algorithm, while communities refer to the potential social structure represented by these clusters. These clusters suggest that these communities of packs tend to recommend the same few accounts that appear in the other SPs. At the same time, these connections between clusters indicate that the network is not completely fragmented, meaning that some accounts appear across multiple groups. There are also more sparsely connected nodes around the outside, which may reflect smaller or more niche packs with less overlap across the broader network.

Overall, these results suggest that SPs do share overlapping accounts that form clearly identifiable communities. However, cross-community connections here indicate that the ecosystem is not fully siloed, meaning that the network graph does not exhibit strongly isolated echo chambers.

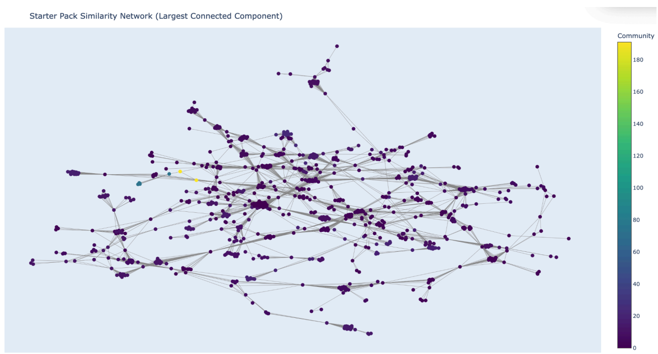

Figure 1: Network visualization of starter packs connected by shared users.

Thematic Categorization of Starter Packs

Our second research question explores whether starter pack descriptions reveal distinct thematic communities that correspond to networks of users with shared interests or professional affiliations. Regarding thematic categorization, our hypothesis is that starter packs are organized with the purpose of forming shared interest and preferences communities instead of a random selection. In order to prove that starter packs are not likely to contain a mix of unrelated accounts but instead community based decisions, we performed a two-stage thematic analysis of starter pack descriptions. In the first stage, we want to determine whether the description can be categorized into a board and generic theme through keyword matching. In the second stage, our goal is to determine whether descriptions reveal recognizable communities, and whether these communities align with the network structure identified through network clustering.

Keyword-Based Theme Identification

As the initial step, starter pack descriptions were classified using a predefined keyword dictionary covering 20 most common online themes such as art, sport, gaming, and news. This simple rule-based text classification method gives us a better understanding of the thematic structure of our data set. By assigning each starter pack to a category based on the keywords found in its name and description, we are able to categorize themes across starter packs and their approximation frequency counts. We did analysis on 1000 random starter packs with description and found that the most frequently identified themes were art, animals, technology, and politics. This initial analysis shows that starterpacks are predominantly created around specific topical interests rather than arbitrary collection of accounts. However, it is also very important to note that 132 starter packs cannot be classified, which highlights the limitation of keyword-based approaches and motivated the use of a more advanced method.

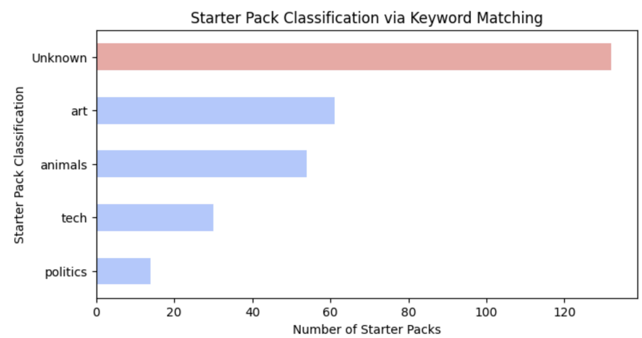

Figure 2: Categories of 1000 randomly sampled starter packs.

Semantic Clustering via Embeddings

To capture a deeper thematic relationship beyond simple keyword matching, SP descriptions were converted into vector embedding using the all-mpnet-base-v2 model. Description will be turned into 768 dimensions dense vector space and will be used for communities detection and clusterings. Since descriptions with similar meaning will occupy nearly positions in vector space, it allows us to observe whether start packs naturally cluster together or are distributed without clear structure.

To cluster them, we initially experimented with density-based clustering methods such as DBSCAN (2024), which groups description vector points by identifying dense regions in vector space. The distance between points are measured through Euclidean distance between embedding vectors, in other words, description vectors with similar meaning will have closer distance. However, this method produced unstable results, with a large number of descriptions around 8,865 out of 10,000 starter packs described fall under cluster -1 - clusters labeled as noise. Additionally, choosing appropriate density parameters(min_cluster_size=10, min_samples=2) is difficult due to the varying densities level of semantic embeddings. Also, when looking at the graph produced by this methodology, a lot of cluster points appear to be elongated instead of spherical form, which density methodology can capture well.

To address these limitations, we constructed a k-nearest neighbor (KNN) graph using cosine similarity and applied the Leiden community detection algorithm (Agashe, 2024) to identify clusters. We first find the k most similar other starter packs with cosine similarity to keep the local semantic neighborhood around each pack. After knowing each pack's nearest neighbors, we build a relationship network with KNN where:

- Node: one starter pack
- Edge: Connect two starter packs are very similar, which mean if pack A is among pack's B's nearest neighbors, we connect them
- Edge weight: Cosine similarity between the two packs

The reason behind computing KNN and building the KNN graph before applying Leiden is we want to avoid computing similarity scores for every single pair, which will result in $O(n^2)$ computation. After having the relationship network, we apply the Leiden community detection algorithm to this graph to identify densely connected groups of starter packs. The Leiden algorithm tries to maximize modularity where it supports splitting solutions where there are many edges inside a community and relatively few edges between communities. For example, if artist-related descriptions are all close in embedding space, KNN graphs will capture this relationship and they will all connect to each other and form a dense region of connection. Leiden will see this dense region and cluster them together as a community. Therefore, we can conclude that these communities represent clusters of starterpacks that share similar thematic content. As a result, the Leiden algorithm identified 19 distinct communities, which correspond to groups of starter packs describing related topics or interests.

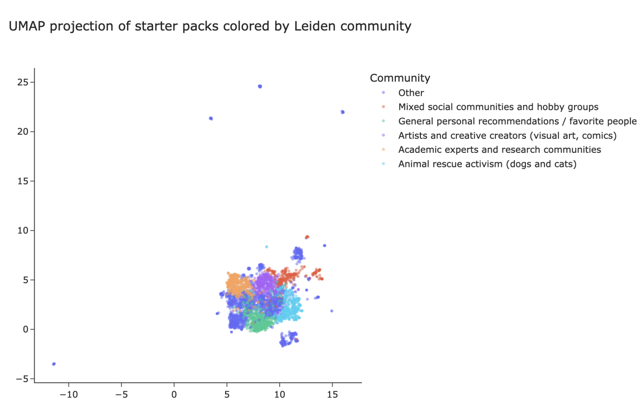

Figure 3: This figure shows top 5 communities within 19 distinct communities as the result of Leiden community detection clustering, where points with similar color form a community. UMAP (2018) is being used to reduce the dimension for 2D visualization.

Following the clustering, we manually reviewed the descriptions within each cluster, with the assistance of OpenAI's GPT-5.2, we assigned labels to identify the dominant theme represented by each community. With deeper analysis, The most popular community type of starter packs is personal recommendation and preferences. This community uses recurring language such as "my favourite" or "people I follow" in their description, which suggest a curated expression of individual taste rather than neutral topic lists.

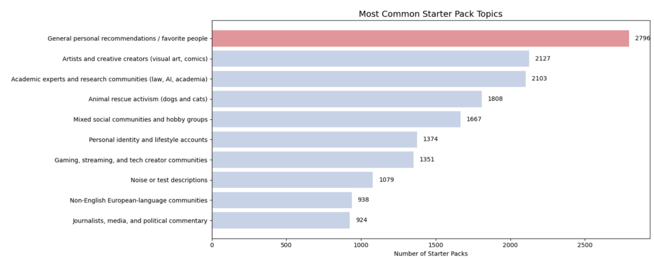

Figure 4: Bar chart of most common Starter Packs topics

In conclusion, packs sharing similar descriptions frequently appeared within the same network clusters. Some communities are strongly associated with a single theme, especially art, animal rescue activism (dogs and cats), and journalism, media, and political commentary. This pattern suggests that the thematic organization of starter packs is not incidental and users consistently group accounts around shared interests, preferences, and social identities. These findings support the view that starter packs potentially expose users to a concentrated set of voices and perspectives with specific themes and personal preferences. This is one of the signs of echo chamber formation.

Starter Pack Popularity

To examine whether follower influence is distributed evenly across the starterpacks as it might describe the popularity of the starterpacks, we analyzed the median follower count of accounts included within each pack and across communities. We chose to compute median follower instead of mean within each pack to avoid heavy-tailed distribution and extreme outliers.

Overall Follower Distribution

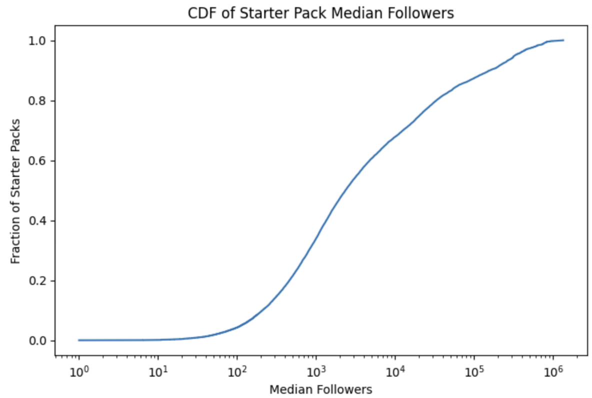

Figure 5: Distribution of median followers in starter packs

Figure 5 describes the cumulative distribution of SPs median followers, which is heavily skewed. In detail, around 85% of starter packs have median follower count below 100,000, which entails that the vast majority of packs are a collection of small or niche accounts. Only a small fraction of starter packs contain accounts with very large followers, which suggests that high influence accounts are the outliers across the starterpacks ecosystem.

Follower Influence Across Communities

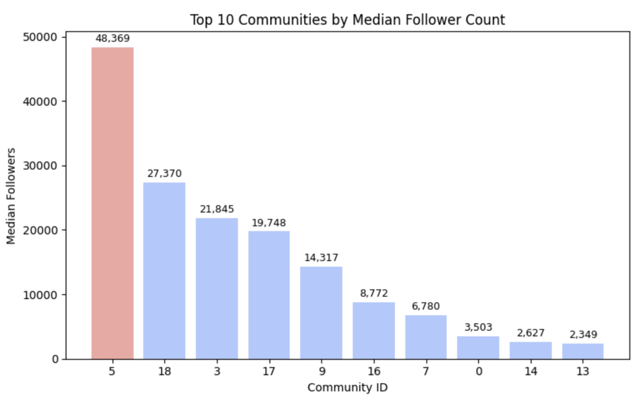

Figure 6: Median followers across communities

We later on combined thematic categorization with starter pack popularity analysis to analyze the influence level of each community. The follower count varies across communities and some communities captured more attention compared to others with higher median followers count. Notably, community 5 (Personality and Lifestyle category) stands out with a median follower count of 48,369, followers by community 18 (Spam and Promotional Accounts) and community 3 (Animal Rescue Activism). The remaining communities show significantly lower median follower counts with 13 out of 19 communities falling under 5,000 median followers. This suggests that only a small number of communities and themes influence the whole starter pack ecosystem. In other words, only some communities attract most of the attention and the distribution is not evenly spread.

Implications on Echo Chamber Dynamics

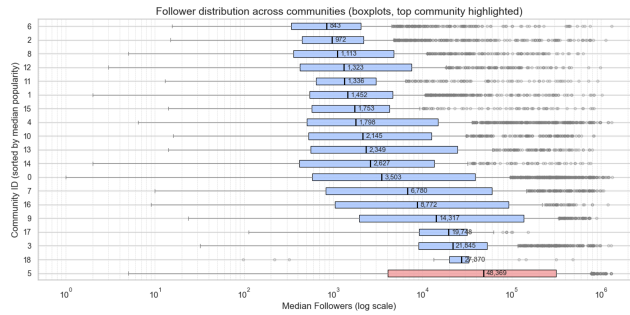

Figure 7: Follower distribution across communities

The uneven distribution of follower influence across communities and starter packs reinforces the notion of central-hub accounts within the ecosystem. Every community also contains a wide range of follower counts (Figure 7) and even in low median follower communities, there are numerous high-follower account outliers. This suggests that influential accounts exist across thematic communities rather than just exist within dominant communities. In conclusion, users in starterpacks are likely to encounter either the same influential voices within each starter pack or the same influential voice of the same thematic starter pack.

Discussion

From our research, we found that while some overlapping accounts exist across different SPs, they remain largely clustered rather than fully mixed, suggesting possible echo-chamber dynamics within groups. Based on this finding, our results also suggest that similar accounts with more followers across communities may potentially have more influence on users following these topics and enhance the echo chambers, since those accounts could have the power to lead the discussion of certain topics.

However, we focused heavily on the account information, such as which SPs the accounts were in and the number of followers, to determine whether they have influence on discussions among communities. In addition, our similarity measurement was mainly based on Jaccard similarity of overlapping accounts across SPs, which focuses on shared account composition rather than actual interaction or information exchange between users. This method has the bias that some accounts could have few posts but a high number of followers, which may create little to no effect on the communities. To further test our hypothesis, we should look into the accounts that we identified as highly influential by examining their posts, interactions in the comment sections, and the number of reposts and likes to better determine their influence.

The individual leaning scores can be referred to in Cinelli et al. 's research (2021), where they computed user leaning based on the content shared by users to identify their impact on certain topics. For our future research, we could also adopt a similar approach to estimate the ideological leaning of accounts within SPs and further examine whether these accounts play a role in reinforcing echo chambers.

Conclusion

At the beginning of our report we defined echo chambers as 1.) mass exposure to the same or similar people, 2.) exposure to the same or similar content, 3.) exposure to influential perspectives within tightly connected communities. Our analysis of user overlap found that, while some overlap exists across communities, they remain largely clustered rather than fully mixed, suggesting possible echo-chamber dynamics within groups. When identifying thematic categories, we discovered that there were 19 distinct topic communities, which separate users into 19 topical groups. Finally, finding SP popularity, some communities tend to contain accounts with higher median follower counts. Thus leading us to conclude that influential accounts within these clusters are the primary spokesperson and central hub within their community(ies). While our findings may be sparse, there is strong evidence that suggests there is some sort of an echo chamber cast throughout BlueSky. For future iterations, we could look further into data about individual accounts within SPs rather than scrapping the top-level.

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